Shashi Kumar Banoth

India
shashikumarbanoth91@gmail.com | +91 9182805541 | linkedin.com/in/shashi-kumar-banoth | github.com/shashi-kumar91 | leetcode.com/u/shashikumar123

Summary

AI/ML Engineer Intern at Samsung R&D and independent founder with hands-on experience building end-to-end production systems across deep learning, LLMs, computer vision, browser automation, and big data engineering. Proven track record deploying real-time face recognition at scale, shipping a production Chrome extension across 18+ ATS platforms, cutting API costs by 40%, and shipping models 8x lighter than published baselines. Adept at orchestrating PySpark and Kafka ETL pipelines and translating complex analytical metrics into stakeholder reports. Seeking roles in Machine Learning Engineering, Data Engineering, AI/NLP Engineering, or Data Science.

Key Achievements

- Shipped a production Chrome extension automating job applications across 18+ ATS platforms
- 19% fewer collisions than published autonomous-driving baselines
- 40% API cost reduction via LLM routing
- 95% face recognition accuracy in a Samsung live production system
- 94.8% AUC on large-scale data classification
- 32% faster inference on edge-deployed YOLOv8
- 8x lighter model (8M vs. 67M parameters) than published baseline

Education

Indian Institute of Information Technology, Design and Manufacturing, Kurnool *2022 – 2026*
B.Tech, Artificial Intelligence and Data Science, May 2026, GPA: 8.12 / 10
Achievements: Solved 200+ DSA problems on LeetCode; active contributor to college ML/AI club. **Coursework:** Linear Algebra, Probability & Statistics, Artificial Intelligence, Data Analysis & Visualization, Software Engineering, DBMS

Skills

- **Domains:** Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Large Language Models, Generative AI, Data Science, Data Analytics, Data Engineering, Computer Vision, MLOps, Browser Automation
- **Machine Learning:** Scikit-learn, XGBoost, supervised learning, unsupervised learning, classification, regression, clustering, model optimization, hyperparameter tuning, statistical modeling, feature engineering
- **Deep Learning:** PyTorch, TensorFlow, Hugging Face Transformers, GATv2, GRU, CNN, BERT, transfer learning, quantization-aware training, ONNX export, edge deployment, inference optimization
- **NLP and LLMs:** LangChain, LlamaIndex, RAG pipelines, FAISS, Sentence-BERT, LoRA, PEFT fine-tuning, LLM API integration, OpenAI API, DeepSeek API, LLaMA, Groq API, prompt engineering, named entity recognition, semantic search, text classification, sentiment analysis
- **Data Science and Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI, exploratory data analysis, statistical analysis, predictive modeling, KPI design, dashboarding, A/B testing, business intelligence, data storytelling
- **Data Engineering:** Apache Spark, PySpark, Kafka, HDFS, Apache Airflow, PostgreSQL, NoSQL, ETL pipelines, ELT pipelines, real-time streaming, batch processing, data modeling, data warehouse, data lake, workflow orchestration, logging, monitoring
- **Computer Vision:** OpenCV, YOLOv8, EfficientNet, MediaPipe, FaceNet, ArcFace, ONNX Runtime, object detection, image classification, real-time video processing, face recognition
- **Web and Browser Automation:** TypeScript, JavaScript, React, Tailwind CSS, WXT, Chrome Extension development, Manifest V3, Shadow DOM, state machine design, Playwright, Chrome DevTools Protocol
- **MLOps, Cloud and Tools:** MLflow, Docker, Git, GitHub, GitHub Actions, Azure ML, AWS SageMaker, Google Colab, Jupyter, VS Code, model versioning, experiment tracking, model monitoring, model deployment, CI/CD, REST API deployment, Google Workspace
- **Programming Languages:** Python, TypeScript, JavaScript, C, C++, SQL, R
- **Core CS:** Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, DBMS

Experience

Founder and Sole Developer, Super Job Agent, Hyderabad, India (Solo, bootstrapped) *Jun 2026 – Present*
- Shipped Super Job Agent, a production Chrome Manifest V3 extension automating end-to-end job applications on LinkedIn Easy Apply and 18+ leading ATS platforms including Greenhouse, Lever, iCIMS, BambooHR, Paylocity, CareerPlug, Hireology, and Oracle Cloud HCM.
- Built the full automation engine in TypeScript, React, Tailwind CSS, and WXT: smart form detection, a state-machine autopilot for multi-step applications with resume upload and submit-preview, per-site adapters for each ATS, a Shadow DOM side panel, and safe new-tab navigation, validated with passing Playwright and Chrome DevTools Protocol end-to-end tests.

- Integrated an LLM-powered answer engine grounded on the candidate's own resume to answer open-ended application questions, with a needs-review fallback for uncertain answers. Smart model routing cut API costs by 40 percent without reducing answer quality.
- Built reliable infrastructure around the product, including CI/CD via GitHub Actions, automated Playwright regression tests to prevent ATS adapter regressions, defensive idempotency guards against duplicate UI mounts, and structured logging and telemetry for autofill events and error reporting, with a low-cost cloud deployment path planned for end users.

AI/ML Engineer Intern, Samsung Research and Development, Bengaluru *Nov 2025 – Apr 2026*

- Engineered and deployed a live production system for real-time face recognition using OpenCV and ONNX Runtime, serving users across the Samsung campus and achieving 95 percent accuracy on 120+ real-world test samples.
- Owned the full ML pipeline with structured logging and observability: extracting 512-dimension face embeddings, running similarity search against a live database, and orchestrating validation across 3 connected subsystems.
- Built 4 anomaly detection modules with adaptive thresholds and behavioral scoring to flag spoofing attempts; streamlined the end-to-end data workflow from raw camera frames to structured CSV and JSON stakeholder reports.

Projects

AI-Powered Safe Route Planning for Autonomous Vehicles – GitHub *Oct 2025 – Apr 2026*

- Trained an autonomous driving model on 850 nuScenes scenes predicting 6 future trajectories via a physics-based bicycle model and occupancy grid scoring, achieving 19 percent fewer collisions than baselines at 8x lower parameter count (8M vs. 67M), training on one GPU in 7.5 hours.
- Leveraged a GATv2 graph neural network to model agent interactions; adversarial noise augmentation improved robustness by 33 percent against sensor noise. Technologies: Python, PyTorch, EfficientNet-B0, GATv2, nuScenes, OpenCV.

LLM-Based RAG Recommendation System – GitHub *Jun 2025 – Nov 2025*

- Built a recommendation engine with LLM API integration (DeepSeek, LLaMA 3.1) that retrieves candidates from a FAISS vector index in under 50ms and re-ranks with robust error handling and retry logic.
- Cost-aware routing slashed API costs by 40 percent by directing simple queries to a lightweight model and complex ones to LLaMA 3.1; a confidence scorer with structured alerting filters uncertain outputs. Technologies: Python, LangChain, FAISS, Sentence-BERT, DeepSeek API, LLaMA 3.1, PyTorch.

Big Data Classification Using SBOA-OGRU – GitHub *Jan 2025 – Apr 2025*

- Processed large-scale data through a PySpark and Kafka architecture: automated feature selection cut dimensions by 35 percent, and the final model hit 94.8 percent AUC after benchmarking multiple architectures. Technologies: Apache Spark, Kafka, PySpark, TensorFlow, HDFS, GRU.

Low-Latency Real-Time Object Detection for Edge Deployment – GitHub *Aug 2024 – Sep 2024*

- Compressed YOLOv8 to one-quarter its size via quantization-aware training (PyTorch to ONNX to TFLite), achieving 32 percent faster inference with negligible accuracy drop. Technologies: Python, YOLOv8, PyTorch, ONNX, TFLite, OpenCV.

HR Automation Dashboard, Python-Driven KPI and Reporting Engine – GitHub *May 2025 – May 2026*

- Built a 12-tab Excel dashboard tracking 100 employees across India and the US with 10 live KPI cards and a 3-layer dynamic alerting system; orchestrated the full data workflow (Python, openpyxl) from raw HR data ingestion to stakeholder-ready PDF and CSV reports with zero manual intervention. Technologies: Python, openpyxl, Excel, VBA.

Real-Time AI Voice Assistant with LLM-Powered Conversational Stack – GitHub *Aug 2025 – Sep 2025*

- Shipped a low-latency speech-to-text to LLM to text-to-speech system (faster-whisper, LLaMA 3-70B via Groq, gTTS) with live interrupt support, amplitude-based silence detection, and a dual CLI plus Streamlit interface; an OpenAI-compatible abstraction enables zero-code LLM provider swapping. Technologies: Python, faster-whisper, LLaMA 3, Groq API, Streamlit, PyAudio.

Certifications

- Data Science Masters 2.0, PW Skills (350+ hours: Python, Machine Learning, Deep Learning, NLP, Computer Vision, deployment) – Certificate
- Data Structures and Algorithms in C++, PW Skills (dynamic programming, recursion, algorithmic problem-solving) – Certificate